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High Resolution Rovibronic Spectroscopy of I_2

RESEARCH PROJECT

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Abstract

A Measured Active Rotational-Vibrational Energy Levels (MARVEL) analysis is performed on $72\,934$ rotationally resolved transitions of I_2 in the literature. The resulting spectroscopic network connected up with 23 calculated "magic number" transitions consists of $71\,174$ validated unique transitions involving $19\,674$ energy levels. The resulting empirical energy levels and associated uncertainties include the $X\,^1\Sigma_{0g}^+$ ground state, the $A1_u$ and $B0_u^+$ valence states, and the $D0_u^+$, $F'0_u^+$, $E0_g^+$, and 1_g ion-pair states. The comparative lack of fluorescence spectrum data compared to that of absorption leaves large sections of the higher vibrational levels of the $X0_u^+$ ground electronic electronic state missing from the spectroscopic network. This also resulted in the need for many calculated magic number transitions to connect up floating networks resulting from fluorescence spectrum data. Comparisons between sources highlights overestimated measurement uncertainties and suggests some calibration errors. Inclusion of other valence and ion-pair states is discussed should data be available. There is a considerable quantity of unassigned high resolution spectra, and assigning this is a logical next step.

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1 Introduction

Molecular iodine is a viable alternative propellant for spacecraft electric propulsion (EP) systems, it is both cheaper to produce and easier to store than the most commonly used propellant, xenon [1]. In an electric propulsion system the propellant is ionised and then accelerated by electrothermal, electrostatic, or electromagnetic methods and ejected from the engine to produce thrust [2]. Both atomic iodine and I_2 have low ionisation potentials and high mass which contribute to to a better thrust-to-power ratio for an EP system [3, 4]. Iodine is a solid at room temperature and is therefore easier to store than xenon gas, and at the amounts needed for propellant usage can be procured for less than 1/20 of the cost of the same amount of xenon [1]. Iodine is unique amongst the contenders for EP propellants in being molecular; this is a downside because energy is lost in exciting molecules into vibration and rotation modes instead of going into the necessary ionisation, but the benefits of iodine outweigh this general drawback of molecules [2]. An in-space demonstration of an iodine propulsion system was performed in 2021, with manoeuvres confirmed with satellite tracking data [1]. It was demonstrated that the use of pressurised propellant enabled a smaller system design. A propulsion system miniaturised enough to be used by the growing number of small satellites would ensure that these satellites are able to avoid collisions and de-orbit safely at their end of life, an important factor in mitigating the long term issue of space debris build up [1].

Improvements for iodine propulsion systems require extensive modelling of the behaviour of iodine plasmas, which in turn require, amongst other things, extensive and accurate data on reaction cross-sections and absorption spectra for both atomic and molecular iodine [4–6]. In [4], absorption spectra were used to track the densities of neutral atomic and molecular iodine in a plasma over various temperatures and pressures. It is partly the molecular nature of iodine which increases its ionisation costs at high pressures which affects its performance as a propellant [5]. It is these studies which motivate further analysis of existing spectroscopic studies of iodine, and the production of accurate vibration-rotation energy levels.

Independently of the use of iodine as a propellant, or of its well known medical and biological significance, the spectrum of iodine is of note because of its extensive use as a frequency standard in the infrared, visible, and near ultraviolet range. Spectroscopic

investigations of molecular iodine for this purpose generally fall into two categories. One is broad atlases covering a large bandwidth [7–13]; a purpose for which iodine is suited due to the dense absorption spectrum of the B-X system in the visible range which contains multiple transitions per cm^{-1} over the whole visible region. These studies, particularly the Fourier-transform measurements conducted at the Laboratoire Aimé Cotton by Luc and Gerstenkorn, are commonly used in spectroscopic investigations of other systems for wavelength calibration.

The other category of study are higher resolution hyperfine-resolved studies of specific rovibrational lines, in order for the hyperfine components to be used as frequency stabilisers for lasers [14–56]. The use of iodine for this purpose is driven by the proximity in frequency of certain transitions with commonly used lasers. In 1969, Hanes and Dahlstrom [14] discovered a coincidence of hyperfine components of the R(127) line of the (11-5) band of the I_2 B-X transition with the radiation of a $633\,\mathrm{nm}$ helium-neon (He-Ne) laser. The authors proposed that these components could be used for laser frequency stabilisation. Around the 532 nm wavelength region, a frequency-doubled Nd:YAG can be tuned to multiple hyperfine resolved iodine lines to allow for calibration [23, 30, 37]. The portability of iodine based laser stabilisation equipment has led to its implementation for laser stabilisation in space [57], also using transitions around 532 nm. Iodine spectra components at both $532\,\mathrm{nm}$ and $633\,\mathrm{nm}$, used with the aforementioned lasers, form part of the recommended radiations for frequency stabilisation for the practical realisation of the metre [58, 59].

The purpose of this study is to start to connect these two areas of interest, the use of iodine for frequency calibration and the need for spectroscopic information for the purposes of modelling iodine plasmas for EP systems. The MARVEL process put forward by Furtenbacher *et al.* [60] is an example of a process of connecting a measured quantity, transition frequencies, with a quantity not directly measurable but of interest on theoretical or practical grounds, in this case a network of rovibrational energy levels and associated uncertainties. The next steps would be to use these energy levels and existing potential energy curves to produce a line list database using the software Duo [61, 62].

The following section introduces the theory necessary for understanding rovibronic transitions in iodine. The choice of quantum numbers used to uniquely describe the MARVEL energy levels is discussed, including Λ-type doubling, and the use of the Ω

representation for electronic states. Selection rules for the relevant single-photon, two-photon, and Raman transitions are discussed, as well as the spectroscopic constants for iodine available in the literature. Section 2.3 summarises the MARVEL methodology: the matrix inversion of the transition file and the use of magic numbers to connect spectroscopic networks. Section 3 discusses the resulting spectroscopic network and energy levels produced in this study, detailing the levels obtained for each electronic state. For purposes of repeatability, sources which have been excluded or where adjustments were made are outlined. A more complete outline of the MARVEL sources is provided in the Appendix.

2 Background

2.1 Relevant States and Quantum Numbers

In order to characterise the spectrum of iodine it is necessary first to describe the relevant electronic states and the choice of quantum numbers to label both electronic states and the rotational, vibrational, and parity sublevels. In the following discussion, vectors and symmetry operators are denoted by bold and italicised letters, bold and non-italicised letters correspond to non-symmetry operators, and non-bold non-italicised letters correspond to scalars and quantum numbers.

The three sources of angular momentum in a diatomic molecule ignoring nuclear spin: the electron spin $\boldsymbol{S}$, electron orbital angular momentum $\boldsymbol{L}$, and the rotational angular momentum of the nuclear framework $\boldsymbol{R}$, couple together to result in a total angular momentum $\boldsymbol{J}$ [63]. The exact way in which the angular momenta couple depends on the relevant Hund's case which varies between molecules and between electronic states. A "good" quantum number is the state label of an eigenbasis which diagonalises the relevant Hamiltonian, and the quantum number is related to the eigenvalues of the relevant Hamiltonian with the choice of eigenbasis. The different Hund's cases correspond to different choices for the Hamiltonian to be diagonalised by the basis functions, for case (a) this is the electronic and rotational Hamiltonians $\mathbf{H} = \mathbf{H}^{el} + B\mathbf{J}^2$, and for case (c) this includes a spin-orbit term $\mathbf{H}^{\mathbf{SO}}$. The good basis functions are simultaneous eigenfunctions of this part of the total Hamiltonian, and of a set of mutually commuting angular momentum operators (the choice of which will

again depend on the choice of Hamiltonian). Take for example a set of angular momentum basis functions labelled $|J\Omega M_J\rangle$, the three state labels are the quantum numbers and are related to the eigenvalues of different angular momentum operators on this basis function. The operator $\mathbf{J}^2$ (the magnitude of the total angular momentum $\boldsymbol{J}$) has eigenvalue $\hbar^2 J(J+1)$ with this eigenfunction. The projection of $\boldsymbol{J}$ onto the internuclear axis, $\mathbf{J}_z$, has eigenvalue $\hbar\Omega$, and the projection of $\boldsymbol{J}$ onto the laboratory fixed Z axis, $\mathbf{J}_Z$, has eigenvalue $\hbar M_J$. J, Ω, and M_J are then both quantum numbers and basis state labels [64].

Apart from the higher energy Rydberg states, electronic states of iodine fall into the Hund coupling cases (a) and (c) [65]. Hund's case (a) is the non-relativistic case and is used when the orbital angular momentum $\boldsymbol{L}$ is coupled to the internuclear axis. The orbital angular momentum parallel to this axis is $\hbar\Lambda$, and Λ is a good quantum number. The total angular momentum $\boldsymbol{J}$ has component $\hbar R$ perpendicular to the internuclear axis and component $\hbar\Omega = \hbar(\Lambda \pm \Sigma)$ parallel to this axis, where Σ is the component of electron spin $\mathbf{S}$ along the internuclear axis [63]. The Hund case (a) term symbol is ${}^{2S+1}\Lambda_{g/u}^{+/-}$. The ground electronic state of iodine falls into Hund's case (a) and is written $X^1\Sigma_g^+$ (Σ here denoting $\Lambda = 0$ analogous to SPDF labelling for L in atoms). The +/- parity term denotes whether the electronic wavefunction is symmetric or antisymmetric with respect to a reflection across a plane containing the internuclear axis, this is only specified for $\Lambda = 0(\Sigma)/\Omega = 0$ states due to states with non-zero orbital angular momentum occurring in +/- pairs (see the rotationless parity discussion below) [66]. *g* (gerade) and *u* (ungerade) denote whether the state is symmetric or antisymmetric with respect to a parity transformation of the electronic coordinates about the centre of (charge) symmetry. This arises for any molecule whereby the field created by the nuclei has a centre of charge symmetry [67], which for I_2 as a homonuclear diatomic exists and is the midpoint of the two nuclei. The parity transformation in question is the inversion operator $\hat{\boldsymbol{\imath}}$, which inverts the coordinates of the electrons (but not the nuclei or spins),

$$\hat{\boldsymbol{\imath}}\psi_{el}(x_i, y_i, z_i) = \psi_{el}(-x_i, -y_i, -z_i) = \pm\psi_{el}(x_i, y_i, z_i) \tag{1}$$

where the electronic states corresponding to eigenvalue $+1$ have *g* parity and those with eigenvalue -1 have *u* parity [66].

Due to large spin-orbit effects the other electronic states fall closer to the Hund's case (c) coupling scheme [65]. Without spin-orbit effects the first excited state should be $^3\Pi_u$ [68], but in Hund's case (c) the separate Ω components become separate electronic states, each with its own potential curve [64]. The only good quantum angular momentum quantum numbers for case (c) are J and Ω, and the electronic term symbol used is just $\Omega^{+/-}_{g/u}$, along with a letter state label. In the framework of molecular orbital theory, the electronic configuration of the valence states are written with four numbers $mpqn$, corresponding to the distribution of the 10 open shell electrons across four molecular orbitals $\sigma_g^m \pi_u^p \pi_g^q \sigma_u^n$ [65]. The molecular orbitals are approximated by linear combinations of atomic $5p$ orbitals on each nucleus. σ_g and π_u are bonding orbitals with lower energies than the corresponding $5p$ atomic orbitals, σ_u and π_g are anti-bonding orbitals with higher energies than the corresponding $5p$ atomic orbitals. The ground state X has configuration 2440 and has the deepest potential. The next possible electron configuration in terms of energy is 2431, the energy levels with this configuration are summarised in table 1. The $C^1\Pi_{1u}$ state is separated due to the exchange interaction, the effective force resulting from the exchange anti-symmetry of electrons. This creates a difference in energy between singlet and triplet states. The three Ω components are split due to spin-orbit interactions between themselves and are considered separate case (c) states. The 0^-_u state is split due to spin-orbit interactions with other (higher) configurations, resulting in the 5 electronic states in table 1. Of these, only the $A1_u$ and $B0^+_u$ states are relevant to this MARVEL study due to the weakly bound nature of the B' and C states (leading to bound-free transitions from the ground state [69]), and the forbidden $\Delta\Omega = 2$ transition necessary to access the $A'2_u$ state from the $\Omega = 0$ ground state (see section 2.2.2). The other 18 valence states are weakly bound [65].

Table 1: Electronic states of iodine that have a 2431 electron configuration.

Case (a) Term Symbol	Standard Label[1]
$^3\Pi_{2u}$	A'
$^3\Pi_{1u}$	A
$^3\Pi_{0u+}$	B
$^3\Pi_{0u-}$	B'
$^1\Pi_{1u}$	C or B", see [70].

Iodine also has ion-pair (IP) states with rotationally resolved transitions studied in the literature. Ion-pair states lie higher in energy than valence states are named as such as they correlate with two ions at their dissociation limit, the first tier are lowest in energy correlating with $^1\mathrm{S}$ positive ion, the second with the $^3\mathrm{P}_{2,1,0}$ positive ions and the third with the $^1\mathrm{D}_2$ positive ion [65]. Under consideration in this study are the $D0_u^+$, $F'0_u^+$, $E0_g^+$, and 1_g ion-pair states. Note that the $1g$ state considered here is not shown in Fig. 1 as it belongs to the third tier of IP states, corresponding to the ^{1}D I$^+$ dissociation limit [72], and it has no standard letter label.

A single vibrational quantum number label v [2] is sufficient for iodine because it is a diatomic and therefore has only one vibrational mode. The rotational quantum number is J. Specifying the total parity of the energy levels is also necessary due to an effect called Λ-doubling which causes states with non-zero Ω to split in energy depending on parity. Rotationless, or *e/f*, parity is defined to conveniently absorb the alternation in total parity due to J. The effect of the operator for total symmetry, $\hat{\boldsymbol{I}}$, on the rotational part of the wavefunction ψ_{rot} is [66]:

$$\hat{\boldsymbol{I}}\,\psi_{\text{rot}} = (-1)^{J-\Omega}\,\psi_{\text{rot}}. \tag{2}$$

Rotationless parity *e* or *f* is defined by factoring out the $(-1)^J$ phase factor such that

$$\begin{aligned} \hat{\boldsymbol{I}}\,\psi_{\text{rot}} &= +(-1)^J\,\psi_{\text{rot}} \text{ for } \boldsymbol{e} \\ \hat{\boldsymbol{I}}\,\psi_{\text{rot}} &= -(-1)^J\,\psi_{\text{rot}} \text{ for } \boldsymbol{f}. \end{aligned} \tag{3}$$

The rotational levels of $\Omega = 0$ states do not have *e/f* parity sublevels: 0^+ states have all *e* parity rotational levels and 0^- states have all *f* parity rotational levels [73].

The final set of four quantum numbers to uniquely describe all the energy levels in this MARVEL process is then:

1. The electronic state term: a string containing the letter label and Ω. The 1_g state includes the gerade symmetry as there is no standard letter label.

[1] The standard practice of using uppercase letters for states with the same spin multiplicity as the ground state and lowercase letters for states with different multiplicity [71] is not used for iodine. The standard state labels in the literature use uppercase letters with primes and double primes, as well as lowercase letters and Greek letters.

[2] Not to be confused with the Greek letter ν for frequency or $\bar{\nu}$ for wavenumbers (energy in units of cm^{-1}).

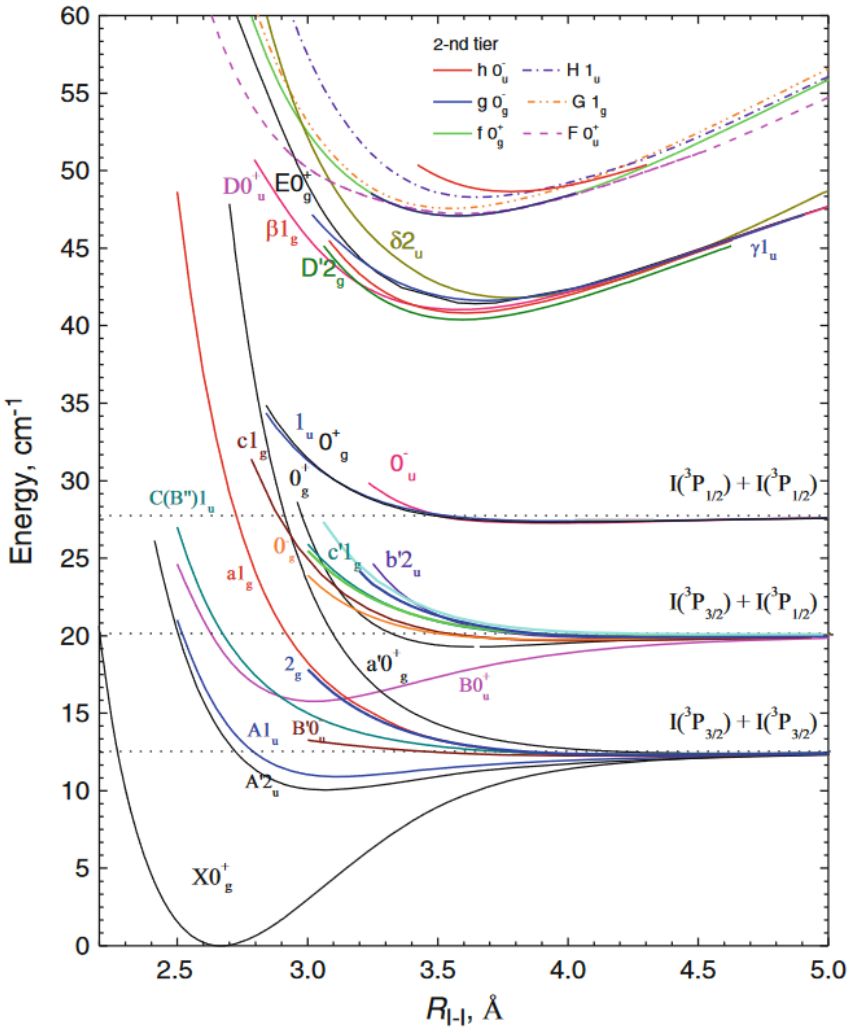


Figure 1: Potential energy curves of the electronic states of iodine from "*The Iodine Molecule*", a useful reference work by Lukashov *et al.* [65]. Note that the y-axis units are incorrect, they should be in units of $1000\,\mathrm{cm}^{-1}$.

2. The rotationless parity *e* or *f*.

3. The vibrational quantum number v.

4. The rotational quantum number J.

2.2 Rovibronic Transitions

2.2.1 Spectroscopic Parameters

The quantity measured in spectroscopy is the energy corresponding to the difference between two energy levels:

$$\bar{\nu} = E_{upper} - E_{lower} \tag{4}$$

which can be the energy of one or multiple photons, or the difference between photon energies. $\bar{\nu}$ is the wavenumber in cm^{-1}, which is the inverse of the vacuum wavelength λ_{vacuum}. The energy in Joules is $E = hc\bar{\nu}$, where h is the Planck constant and c is the speed of light in $\mathrm{cm\,s}^{-1}$. The frequency or wavelength of photons is also often used instead of the wavenumber, depending on the purpose of the experiment and the region of the EM spectrum.

An *a priori* prediction of these energy levels requires solving for the exact wavefunction ψ_i from the time-independent Schrödinger equation for each level *i*:

$$\mathbf{H}\psi_i = E_i\psi_i \tag{5}$$

where **H** is the total Hamiltonian. It is useful in solving this equation to treat the nuclear and electronic motions as separate as they happen at different time scales, known as the Born-Oppenheimer approximation. The wavefunction to solve then is the Born-Oppenheimer product function [64]:

$$\psi^{\text{BO}} = \psi_{\text{el}}\psi_{\text{vibration - rotation}}. \tag{6}$$

To a less good approximation, the vibrational and rotational motions can also be separated [71]. The corresponding eigenvalue, being the total energy considered in

rovibronic spectroscopy can then be written as the sum

$$E^{Total} \approx E^{el} + G(v) + F(J). \tag{7}$$

The terms are the electronic energy (sometimes denoted T), the vibrational energy $G(v)$, and the rotational energy $F(J)$. The separate rotation and vibration energy terms can then be usefully modelled semi-empirically such as the vibration as an anharmonic oscillator:

$$G(v) = \omega_e \left(v + \tfrac{1}{2}\right) - \omega_e x_e \left(v + \tfrac{1}{2}\right)^2 + \omega_e y_e \left(v + \tfrac{1}{2}\right)^3 + \dots \tag{8}$$

where ω_e is the harmonic frequency, and x_e, y_e, and onwards are dimensionless anharmonic correction constants [71]. The rotational energy of a given vibrational level can be modelled as a non-rigid rotor [66]:

$$\begin{aligned} F(J) =& B_v J(J+1) - D_v(J(J+1))^2 + H_v(J(J+1))^3 \\ &+ L_v(J(J+1))^4 + M_v(J(J+1))^5 + ... \end{aligned} \tag{9}$$

where B_v is the rotational constant, D_v is the centrifugal distortion constant for a given vibrational level, and H_v, L_v, and M_v are higher order centrifugal distortion corrections. The terms corresponding to higher order $J(J+1)$ correct for the stretching of the bond as the molecule rotates. An expression with just the B_v term assumes the molecule is a rigid rotor. The v subscripts are necessary because the rotational energy changes with vibrational energy. For the rotational constant this is usually parameterised as $B_v = B_e - \alpha_e \left(v + \frac{1}{2}\right) + \dots$ [66], with B_e the equilibrium rotational constant. The comprehensive set of molecular constants of the B and X states of iodine produced by Luc and Gerstenkorn from their Fourier-transform atlas uses the method of a Dunham expansion [74]. The Dunham expansion is a polynomial representation given by [64]:

$$E_v, J = \sum_{l,m} Y_{lm} \left(v + \tfrac{1}{2}\right)^l [J(J+1)]^m. \tag{10}$$

The expansion is a model-free algebraic representation of the expected smooth energy dependence on v and J. The Dunham coefficients Y_{lm} do not have a direct physical meaning but can be related to the standard spectroscopic constants. The authors provide

the rotational constants B_v, centrifugal distortion constants, and vibrational energies of 81 vibrational levels of the $B0_u^+$ state and 20 vibrational levels of the $X0_g^+$ state [74]. The electronic energy is absorbed into the vibrational energy term E_v corresponding to the band origin, which takes into account the vibrational zero-point energy of the two states and the electronic energy of the excited state. Within each vibrational level of the B state the energy is given by $E_v = T_{0,0} + G(v) - G(0)$, where $T_{0,0}$ is the energy of the $(v = 0, J = 0)$ level B state above the $(v = 0, J = 0)$ level of the X state.

The energy of a rovibronic transition can be derived for the P, Q, and R branches corresponding to $\Delta J = -1$, 0, and 1 respectively. Note that the quantum numbers of the upper state are written with single primes, and those of the lower state with either double primes or no prime. The energy corresponding to a rovibronic transition using the approximate energy of Equation (7) is

$$\bar{\nu} = \Delta E = E_{upper}^{el} + G(v'') + F(J'') - [E_{lower}^{el} + G(v') + F(J')]. \tag{11}$$

In terms of the spectroscopic constants from [74] this becomes

$$\begin{aligned} = E_{v''} - E_{v'} + B_{v'}(J') - D_{v'}(J')^2 + H_{v'}(J')^3 + L_{v'}(J')^4 + M_{v'} \\ (J')^5 - [B_{v''}(J'') - D_{v''}(J'')^2 + H_{v''}(J'')^3]. \end{aligned} \tag{12}$$

By setting $\alpha = (J'' + 1)(J'' + 2)$, $\beta = J''(J'' - 1)$, and $\gamma = J''(J'' + 1)$, the transition energies of the P and R branches are

$$\begin{aligned} \bar{\nu}(\mathrm{R}) =& E_{v''} - E_{v'} + B_{v'}\alpha - D_{v'}\alpha^2 + H_{v'}\alpha^3 + L_{v'}\alpha^4 + M_{v'}\alpha^5 \\ & - [B_{v''}\gamma - D_{v''}\gamma^2 + H_{v''}\gamma^3] \\ \bar{\nu}(\mathrm{P}) =& E_{v''} - E_{v'} + B_{v'}\beta - D_{v'}\beta^2 + H_{v'}\beta^3 + L_{v'}\beta^4 + M_{v'}\beta^5 \\ & - [B_{v''}\gamma - D_{v''}\gamma^2 + H_{v''}\gamma^3]. \end{aligned} \tag{13}$$

2.2.2 Selection rules

As a homonuclear diatomic, iodine has no permanent dipole moment, so pure rotational or vibratational transitions are forbidden. The rotation and vibration structure are discerned only through rovibronic and Raman transitions. Within the Hund's case (c) terms, the single-photon electric dipole electronic transitions have the

angular momentum selection rule $\Delta\Omega = 0, \pm 1$ [65]. For homonuclear diatomics a change in gerade parity $g \leftrightarrow u$ is required, which is an equivalent explanation for why there are no vibrotational transitions within an electronic state [71]. Transitions between the $^3\Pi$ spin-orbit states are disallowed as they share u parity.

In principle, any change in vibrational quantum number is allowed for an electronic transition. Vibrational bands are denoted (v'-v''), where v' is the upper state vibrational level and v'' is the lower state vibrational quantum number, regardless of which state has the higher vibrational energy $G(v)$. The occurrence of different bands within an electronic transition are not equally likely, however. They depend on the expected population of the initial vibrational level, which for absorption from the ground X state is discussed in section 3.1.1, and on the Frank-Condon factor (FCF) which is the squared overlap integral of the two vibrational states: $|\langle v'' \mid v' \rangle|^2$ [75]. Classically, the most probable transition corresponds to a "vertical" transition which does not change the internuclear separation because the electronic transition occurs much quicker than the resulting change in the nuclear framework [66]. This is called the Frank-Condon principle and vertical transitions are named as such because the energy difference corresponds to the vertical line between the two vibronic levels on a PEC diagram.

There is a strict $\Delta J = 0, \pm 1$ selection rule for the change in rotational level in electronic transitions, corresponding to the P, Q, and R branches discussed in the previous section. Transitions between the $\Omega = 0$ states must have $\Delta J = \pm 1$ and therefore lack a Q branch. The rotational branches correspond to the required change in rotationless parity:

$e \leftrightarrow f$ for the Q branch

$e \leftrightarrow e$ and $f \leftrightarrow f$ for the P or R branch [73].

Raman spectroscopy studies used in the MARVEL analysis involve only vibration-rotation transitions of the ground $X\,^1\Sigma_{0g}^+$ state. The vibration-rotation Raman selection rules for a $^1\Sigma^+$ diatomic molecule are $\Delta J = 0 \pm 2$ and $\Delta v = \pm 1$ with $\Delta v = \pm 2$ or greater weakly allowed [66]. With resonance Raman spectroscopy, where the initial excitation is on resonance with an electronic transition rather than to a virtual state, Raman transitions within the ground state for $\Delta v = 16$ have been observed [76]. $\Delta J = -2$ transitions are denoted the O branch and $\Delta J = 2$ the S branch.

2.2.3 Ortho and Para Networks

There is an additional symmetry that is important in understanding the spectrum of iodine, that of symmetric/antisymmetric or *s/a* parity. The iodine nuclei with nuclear spin $I = 5/2$ are Fermions, and every half rotation interchanges the identical nuclei. As a result the total wavefunction of the molecule must be antisymmetric with respect to the operator $\hat{P}_{12}$ exchanging the two nuclei. The nuclear component of the total wavefunction can on its own be symmetric or antisymmetric. The only way then to give the correct symmetry for the total wavefunction is by ensuring the rotational state makes the whole wavefunction antisymmetric. So the proportion between the odd J and even J rotational state populations is determined by the statistics of the nuclear symmetry. Using a general rule for a Σ_g^+ molecule derived in [63] for iodine with $I = 5/2$, the proportion of symmetric and anti-symmetric nuclear states in the ground electronic state is:

$$\frac{N_{symmetric}}{N_{antisymmetric}} = \frac{I+1}{I} = \frac{7}{5}$$

So there should be a 7:5 ratio in population between rotational states with odd and even values of J. The resulting variation in (relative) intensity between transitions starting from odd and even J-values is reported by Simmons and Hougen [7] and is clearly visible in the spectrum provided in their paper. The symmetry of the rotational states with respect to nuclei exchange is labelled s for symmetric states and a for antisymmetric states [66].

There is an additional selection rule that $s \leftrightarrow a$ transitions are forbidden becuse electric transitions cannot simultaneously slip nuclear spins [66], thus dividing the spectrum into two distinct networks: *ortho* and *para*. The network denoted *ortho* is the one with the larger statistical weight, so for iodine it is the *s* states or the odd-J in the ground electronic state which form the para network. But the separation of the spectroscopic network into two distinct parts can be explained when the Δ *e/f* selection rules in the previous section are employed. From a given rotational level, when a transition changes J by one, the rotationless parity does not change, and when the transition changes J by an even number either through a Q branch or through two P/R branch transitions the rotationless parity changes. This means it is impossible to connect an odd-J level of the ground state to an even-J level of the ground state through any number of transitions following these rules. There must then be two

distinct networks.

2.3 The MARVEL process

2.3.1 Matrix Inversion

The connection between transitions and energy levels is given by the overdetermined set of linear equations:

$$\mathbf{A}\boldsymbol{x} = \boldsymbol{y}. \tag{14}$$

$\boldsymbol{y}$ is a vector of all the transitions in energy units (dimension N_t where t is the number of unique measured transitions), and $\boldsymbol{x}$ to be solved is a vector of dimension $N_1 - 1$ containing the energy levels. $\mathbf{A}$ is an $N_t \times (N_1 - 1)$ matrix with elements:

$$A_{ij} = \begin{cases} -1 & \text{if } E_j \text{ is the lower level of the } i\text{th transition} \\ 1 & \text{if } E_j \text{ is the upper level of the } i\text{th transition} \\ 0 & \text{otherwise.} \end{cases} \tag{15}$$

This matrix fully describes which energy levels connect via which transitions. The weighted least-squares solution for the vector $\boldsymbol{y}$ can be solved by first defining $\mathbf{A'} = \mathbf{A}^T\mathbf{G}\mathbf{A}$ and $\boldsymbol{b} = \mathbf{A}^T\mathbf{G}\boldsymbol{y}$, where $\mathbf{G}$ is a the weight matrix with elements $1/\delta_{ij}$ where δ_{ij} is the uncertainty of the transition connecting the *i*th and *j*th energy level. The weighted solution is obtained by solving [77]

$$\mathbf{A'}\boldsymbol{x} = \boldsymbol{y} \tag{16}$$

for $\boldsymbol{x}$. The energy uncertainties can be obtained either from the diagonal elements of the inverse of the weighted matrix **A'** or with an additional bootstrap technique. The version of marvel used in this study includes a bootstrapping technique which works by multiplying each provided transition uncertainty by a random number between 1 and 10, and then running the MARVEL process before randomising the uncertainties again, repeating the process by a default of 100 times. If the averaged energies obtained over these repetitions differ from the original energy by a prescribed amount then the uncertainty in the energy is increased [78].

The actual process undertaken centred heavily on preparing the list of transitions for the MARVEL input. The majority of the available transitions were on scanned hard copies and required the use of a character recognition software to obtain the transitions and assignments in text format. Repeated MARVEL runs were required source by source to catch scanning errors.

2.3.2 Magic Numbers

As discussed in section 2.2.3, the *ortho* and *para* networks of iodine are distinct, and this can be seen by running the available transitions through MARVEL, the output consists of two roughly equally sized components. To connect these up into one complete spectroscopic network it is necessary to compute a magic number transition, a transition which is not physically possible but can be calculated easily using the spectroscopic constants discussed previously. The chosen magic number transition was the $J = 0 \rightarrow J = 1$ transition of the (0-0) vibrational band within the $X0_u^+$ ground electronic state. The energy of this transition is just

$$\begin{aligned} \Delta E &= B_{v'=0}J'(J'+1) - B_{v''=0}J''(J''+1) \\ &= 2B_{v'=0} = 0.074\,623\,14\,\mathrm{cm}^{-1}\,[74]. \end{aligned} \tag{17}$$

The centrifugal distortion constant at this number of significant figures is negligible, being around $6 \times 10^{-9}\,\mathrm{cm}^{-1}$.

This is the only strictly necessary magic number but the final transition file includes 27 others. These are to connect floating networks disconnected from the main two networks due to unobserved transitions. The floating networks were connected if they contained more than 12 transitions and could be accessed through artificial X state rovibrational transitions. They were calculated using the ground state spectroscopic constants from [74].

3 MARVEL results

3.1 The Spectroscopic Network and Energy Levels

The resulting spectroscopic network contains $19\,674$ energy levels split across 7 electronic states, each with both an uncertainty estimated directly from the provided transition uncertainties as well as uncertainties calculated from the bootstrap method. Fig. 2 is a plot of every energy level against its J value. The band separation is most visible for the X state: larger spacings correspond to a steeper potential and larger fundamental harmonic ω_e. The energy within each band can be seen to curve up with what should be $J(J+1)$ dependence. Table 2 shows the first five resulting energy levels and their bootstrap uncertainties. The lowest state "X0 e 0 0 0" has both an energy and uncertainty of zero by definition. This close to the first two energy levels there is little empirical information to be gained by comparing adjacent energy levels because their energy difference is due solely to the calculated magic number between the first two. What can be checked is the energy difference between the first two *para* levels. Note that there is no single transition directly connecting these two energy levels. "X0 e 0 2" should be 6 times the equilibrium rotational constant above the ground state. Dividing the MARVEL energy by 6 gives $0.037\,72 \pm 0.000\,60\,\mathrm{cm}^{-1}$, compared to a literature value of $0.037\,31\,\mathrm{cm}^{-1}$ [79]. It is off by $0.0004\,\mathrm{cm}^{-1}$ which is just about under the stated uncertainty.

Table 2: The first 5 MARVEL energy levels

Level	Energy / cm^{-1}	Unc. / cm^{-1}	No. of transitions
X0 e 0 0	0.000000000	0.000e+00	55
X0 e 0 1	0.074623144	1.000e-05	68
X0 e 0 2	0.226300496	3.578e-03	86
X0 e 0 3	0.444538106	6.000e-03	107
X0 e 0 4	0.748834366	6.544e-03	109

3.1.1 The *X* state

From a theoretical standpoint only the magic number connecting the *ortho* and *para* networks should be necessary. With sufficient data there should be no other floating

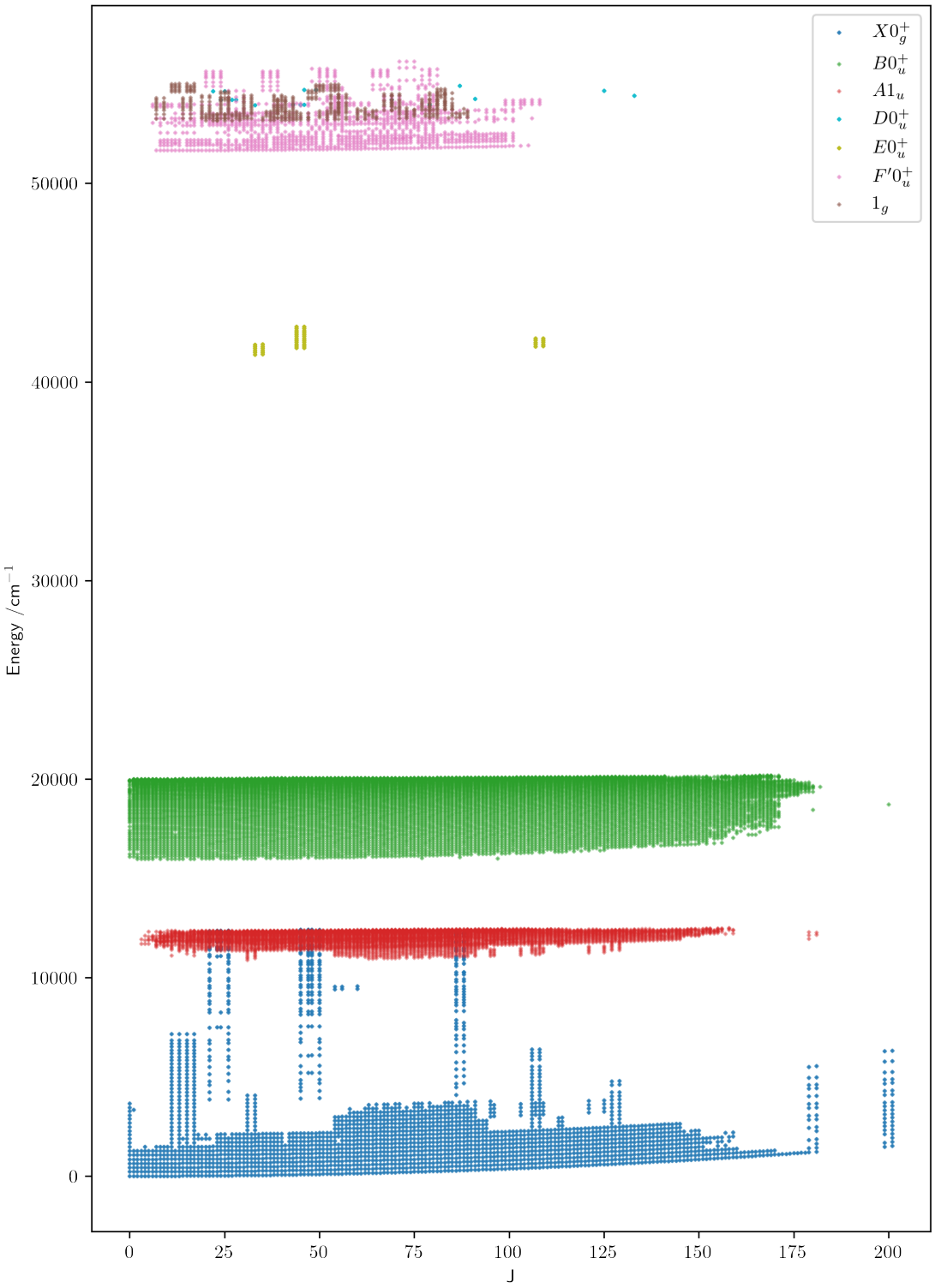


Figure 2: Plot of all the MARVEL energy levels against J

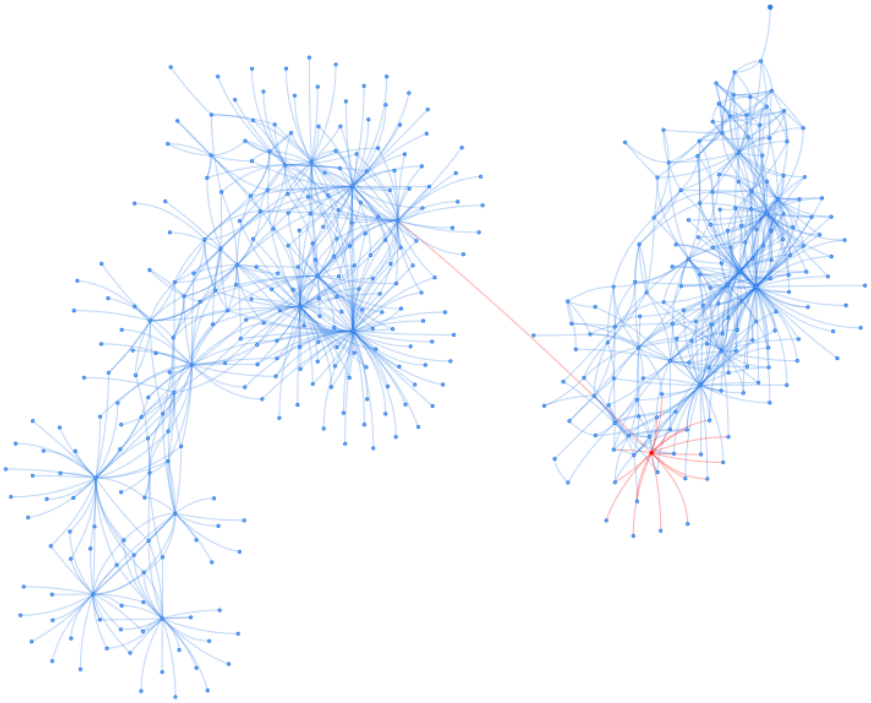


Figure 3: A subgraph of the spectroscopic network showing energy levels within 10 levels of connection of the ground state “X0 e 0 0”, the red node at the bottom right. The two distinct sections are the *ortho* and *para* networks, connected with the magic number MAGIC.1 shown as the red line spanning the networks. Made using a built-in application of the MARVEL 2.1 online version [80]

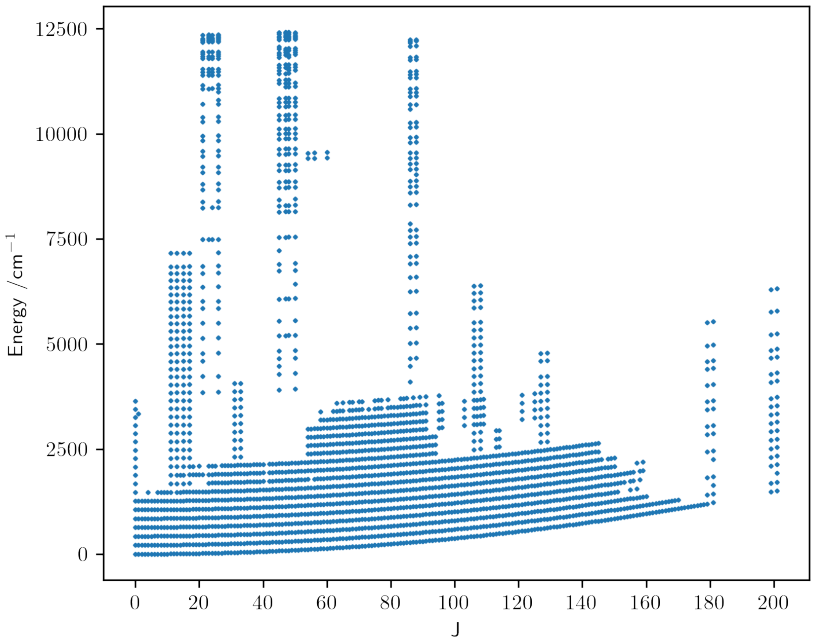


Figure 4: Energy vs J plot for the $X0_g^+$ state energy levels

spectroscopic network. One reason for the sizeable floating networks without magic numbers is the lack of published fluorescence line lists. Fluorescence spectra probe the higher vibrational levels of the ground state more than absorption spectra due to the small Boltzmann factors for high vibrational levels of the ground state. The highest vibrational level of the X state in transitions from the B-X Aimé Cotton atlas is $v = 7$, with only a handful of transitions involving this level. At their highest iodine cell temperature of $250\,^{\circ}\mathrm{C}$ ($523\,\mathrm{K}$) the expected population of this vibrational state relative to the ground vibrational state can be estimated with a Boltzmann distribution with the energy scale K_BT in wavenumber units:

$$\frac{P_{v=7}}{P_{v=0}} = \frac{\exp\left(-G_{v=7}/K_BT\right)}{\exp\left(-G_{v=0}/k_BT\right)} \tag{18}$$

which gives the $v = 7$ state a population 0.022 times that of the ground state. The above calculation for the cell temperature of the B-X portion of the atlas (source 85GeLu [8]) and the cell temperature used for the A-X portion which was later assigned by Appadoo *et al.* (source 96ApLeBeGe [69]) is shown in table 3. The

Table 3: Estimated Boltzmann populations of the X state.

v'	Relative Pop. at $523\,\mathrm{K}$	Relative Pop. at $773\,\mathrm{K}$
0	1	1
1	0.5738717	0.672136391
2	0.3303854	0.452803842
3	0.1908201	0.305747173
4	0.1105689	0.206928482
5	0.0642771	0.140375062
6	0.037489	0.095450541
7	0.0219374	0.065056823
8	0.0128799	0.044446828
9	0.0075874	0.030438951
10	0.0044848	0.020896201
11	0.0026599	0.014380092
12	0.001583	0.009920212
13	0.0009453	0.006860465
14	0.0005665	0.00475629

absorption experiment conducted at $773\,\mathrm{K}$ included transitions from as high as the $v = 14$ level of the X state.

Fig. 4 shows how sparse the upper vibrational levels are in the available data. The levels that are accessed are through fluorescence spectra, in which the upper state is excited with a vertical transition, relaxes non-radiatively, and then decays through emission also vertically but terminating on high v levels of the X state [66]. Fluorescence doublets are visible, the pairs of vertical lines in the plot, corresponding to the P and R branch fluorescence from a single $B0_u^+$ or $D0_u^+$ excited state. Fluorescence from a single excited rotational state can only end up on two J values but on any vibrational level with which the upper state has a large enough overlap integral. There are numerous fluorescence studies characterising the upper vibrational levels but published without the accompanying transition lists [81–84].

3.1.2 The *B* state

The energy vs J plot for the B state shows extremely densely packed energy levels close to the dissociation limit. The issue here is the opposite of that of the X state,

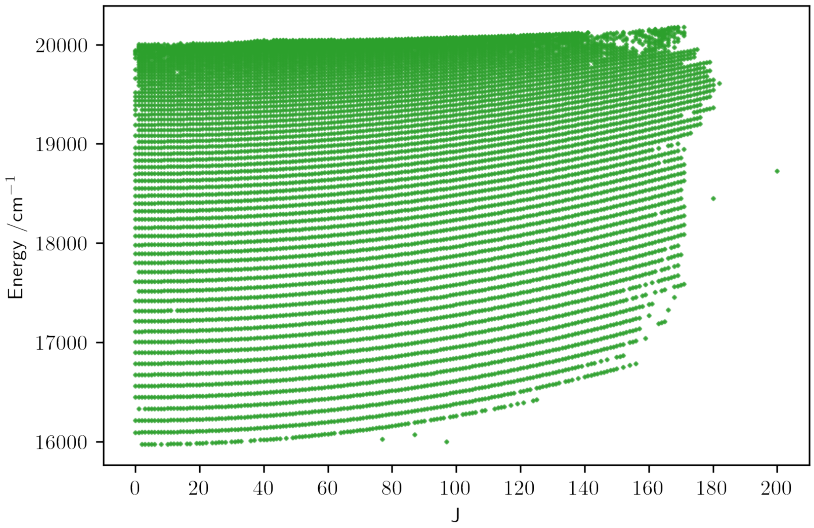


Figure 5: Energy vs J plot for the $B0_u^+$ state energy levels

higher v levels are well characterised due to favourable Frank-Condon factors from the lower v levels of the X state; but lower v levels of the B state are sparse. The three lower energy levels between $J = 70$ and $J = 100$ which look like outliers belong to the first and second vibrational levels. Some of the predicted energies lie above the dissociation energy of $D = 20\,043.063 \pm 0.020\,\mathrm{cm}^{-1}$ [85] by up to $135\,\mathrm{cm}^{-1}$. If correct, this would mean a bound-free absorption, which matches the behaviour of the energy levels at the top right of the plot. The vibrational and rotational structure appears to change completely. The source for the transitions predicting these levels is the Aimé Cotton atlas which provided definite energy levels assignments to these transitions so it is unclear how exactly the bound-free transitions were accounted for in the source.

3.1.3 The *A* state

The levels of the A state were accessed through transitions from only two sources: the assigned lines measured as part of the infrared Aimé Cotton atlas [69] and as an intermediate state to the $1g$ ion pair state [72]. The difference in energy between the *e* and *f* parity levels is not visible in the plot, as the energy splitting is around $0.05\,\mathrm{cm}^{-1}$, scaling with the vibrational level and $J(J-1)$. There are 929 A state levels in the network with both *e* and *f* sublevels, and their energy splitting is plotted in Fig. 7. The

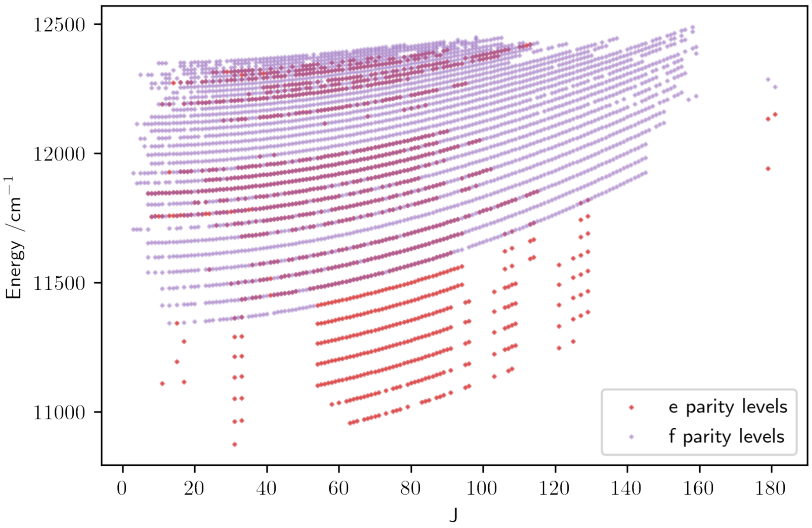


Figure 6: Energy vs J plot for the $A1_u$ state energy levels

Λ-doubling for the *A* state is parameterised by the empirical equation [69]:

$$\pm[Q_{01} + Q_{11}(v + \tfrac{1}{2})][J(J+1) - 1]. \tag{19}$$

The +/- corresponds to deviation of *e* and *f* levels respectively from the rovibronic line centre. The -1 in the rotational term corresponds to $-\Omega$, and the Q parameters are both positive for the A state. Fig. 7 shows this rough relationship which should be a straight line through the origin for each v'. The variation in splitting near the y axis is about $0.025\,\mathrm{cm}^{-1}$ in both directions but this corresponds to $J = 7$ which is the lowest vibrational level with both *e* and *f* parity accessed. The expected difference in energy splitting between the $v = 0$ and $v = 31$ levels for $J = 7$ using the above equation is $6.823 \times 10^{-4}\,\mathrm{cm}^{-1}$ which is still much less than than the $0.05\,\mathrm{cm}^{-1}$ in the plot. Many of the MARVEL energy levels for the A state however have uncertainties of the order $0.1\,\mathrm{cm}^{-1}$ anyway, and when the difference between two levels is calculated the uncertainty propagates with $\sqrt{0.01^2 + 0.1^2}$, meaning this variation around the origin is well within the energy uncertainties.

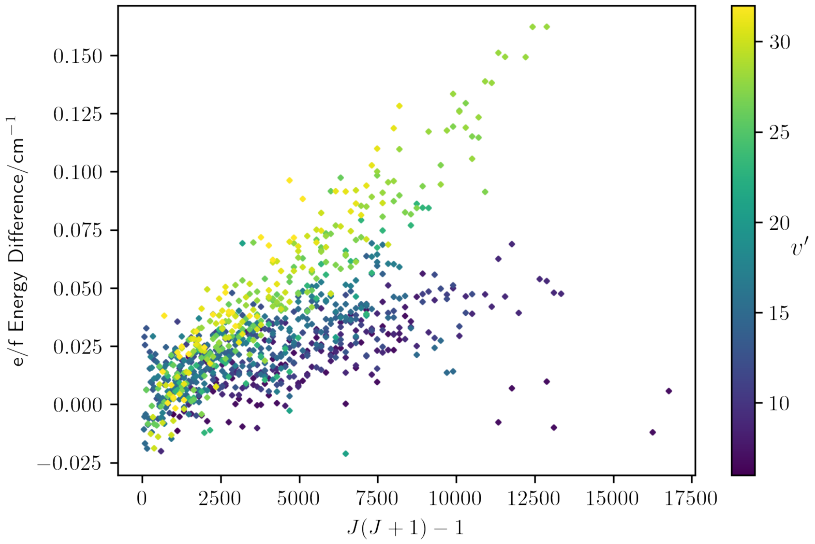


Figure 7: The difference between the energy of e and f parity level energies in the $A1_u$ state

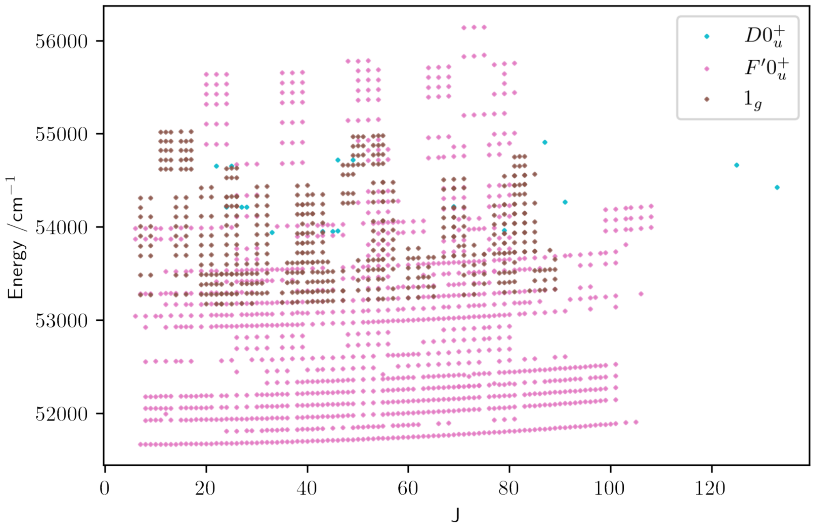


Figure 8: Energy vs J plot for some ion-pair state energy levels

3.1.4 Ion-pair states

The levels of the $E0_u^+$ state are not shown as they lie below the other IP states at around $42\,000\,\mathrm{cm}^{-1}$. The $1g$ is the only state other than the A state which has *e/f* splitting. The assignments in the source (93IsTaOb) [72] did not give the explicit parity assignments but they could be inferred from the two-photon transition assignments given. This is explained in the sources section below. The number of rotationally resolved studies for the ion-pair states is small considering the D-X flourescence has been observed since 1913 [86], much of the data for the D state only provides band positions [87–89]. Data has been missed or spectra are recorded for more IP states, it is recommended that the dissociative positive ion correlating with the IP state is specified as a label due to the large number of IP states and confusing names. Such as some states not having a letter label or there being F, F', f, and f' states. The large number of other valence states are not of concern at the moment due to their dissociative nature.

3.2 Sources

Sources are included in this section if they are extensive and of note, or if adjustments have been made such as applying a correction or omitting a large number of lines. As the most extensive source, the Aimé Cotton iodine atlas first published in 1978 and with assignments in 1985 [8] is first, and subsequent sources both in this section and in the Appendix are ordered chronologically by date of publication.

3.2.1 85GeLu [8]

With over $51\,000$ this was the most extensive source, covering the $14\,800\,\mathrm{cm}^{-1}$ to $2000\,\mathrm{cm}^{-1}$ range. Every transition was part of the B-X system. This source was a repeat of a 1977 atlas [107] with more lines resolved and now fully assigned. It was found that the previous atlas needed to be adjusted by a multiplicative factor of $(1+\delta)$ where δ is written as $-3.8\times10^{-7}\,\mathrm{cm}^{-1}$ even though it follows from the equation that it should be dimensionless. This correction has already been applied to the data given. The atlas totals 571 pages (some repeating in the scan) with two sets of 7 columns per page showing an atlas number, the original atlas and new measured wavenumber, an

Table 4: MARVEL sources used.

Source	Wavenumber range / cm^{-1}	total/used lines
64BrKl [90]	$16956.20000 - 17087.19000$	66/66
69HaBa [15]	$15798.00319 - 15798.00319$	1/1
69StCaWe [91]	$15766.20000 - 15833.82000$	621/432
72ByHeKiLe [92]	$16162.97523 - 16190.72193$	9/9
72KiBea [76]	$417.20000 - 3260.00000$	51/51
72KiBeb [93]	$628.00000 - 1238.00000$	9/9
75PaDwHo [94]	$19429.27000 - 19430.35000$	19/3
77GeLuPe [95]	$18290.56970 - 18697.63580$	323/322
77SiHo [7]	$17999.87400 - 19000.18900$	7806/7799
79BrCrHa [96]	$18901.79600 - 19000.74400$	20/18
79CuLiFrGr [97]	$18232.87022 - 24181.45766$	46/46
79TaDaGi [98]	n/a	105/0
80TeMcSu [99]	$42293.44000 - 50816.53000$	354/354
81BrCrErYa [100]	$16627.85200 - 18298.21000$	12/12
83HlFa [16]	$15233.36744 - 16340.65808$	3/3
85DiPaKhAl [101]	$1452.16000 - 1676.38400$	12/12
85GeLu [8]	$14800.77690 - 19978.79490$	51330/51284
86IsToShTa [102]	$34205.00000 - 36746.40000$	1336/1336
87BrHo [103]	$35826.20000 - 36130.20000$	13/13
92SuMaKl [104]	$16999.13590 - 17001.26100$	35/34
93IsTaOb [72]	$4259.96000 - 42378.31000$	769/588
95JuEiSwYe [23]	$18788.30372 - 18788.30372$	1/1
96ApLeBeGe [69]	$7305.07470 - 11708.04850$	9552/8344
96EdBaGiRo [105]	15703.05444	1/1
97Sansonet [25]	$15233.36744 - 17578.59945$	4/4
98BoBoKnNi [26]	$12601.91524 - 12796.05860$	2/2
98BoKlKnTi [27]	$12262.32285 - 12262.34032$	2/0
99EdBaGiRo [28]	15797.94382	1/1
99YeRoPiLo [30]	$18786.62238 - 18788.40853$	7/7
00BoKlWiKn [31]	$12578.29006 - 12579.73619$	3/0
03Quinn [58]	18788.33868	2/1
04DuTr [38]	$13922.83480 - 13923.57610$	4/3
04EnHaRaWo [39]	$14818.15070 - 14819.36940$	12/11
05BuDaAmGo [41]	$19926.00156 - 19926.00156$	1/1
05Chen [40]	19429.79908	1/1
06ReSaKaNo [42]	17686.31665	1/1
07ReBeGeHo [43]	12962.48787	2/1
09ZhLuWa [46]	17880.33955	1/1
10Williams [106]	$12275.27100 - 19854.48900$	350/349
12YaMeZhPe [47]	17811.455381	3/3
13HsKaChCh [48]	18227.22265	1/1
13HuChChSh [49]	14903.86663	1/1
13ShChChHs [50]	18684.65577	1/1
16KoAkHoIn [52]	$17288.22853 - 17288.79587$	4/4
18HuGuSuSh [53]	15462.60270	1/1
19ChDeLiLi [54]	18726.94218	1/1
19NeIgKuMa [55]	$9360.59390 - 9490.88794$	6/6
24ChXiQiMi [56]	18042.01913	1/1
MAGIC	$0.07462 - 4095.89170$	28/28

arbitrary intensity, assignment, and calculated wavenumber. There are numerous assigned calculated wavenumber with no corresponding measured line, and a few measured lines with no assignments per page.

3.2.2 69StCaWe [91]

The wavenumbers of the 49-1 band were lower than the same transitions in 85GeLu, and by what is predicted from the rest of the MARVEL network, by around $10.0\,\mathrm{cm}^{-1}$. The transitions in the 6-3 band were lower than the atlas by between 0.1 and $0.2\,\mathrm{cm}^{-1}$. The transitions in the 7-3 band were greater than those in the atlas by between 0.04 and $0.1\,\mathrm{cm}^{-1}$. The transitions in the 43-0 band were lower than those in the atlas by between 1 and $2\,\mathrm{m}$. An issue with this band was noted by Yee and Miller [81] without further details. Transitions in the other bands did not show a uniform disagreement with the atlas or other sources. The number of 49-0 band measurements had a sufficient range to show whether the data has a relative or absolute offset, and it can be seen that the offset is around $9.8\,\mathrm{cm}^{-1}$ at the higher wavenumber end and $11.4\,\mathrm{cm}^{-1}$ at the lower wavenumber end. Applying a simple multiplicative calibration fix (in this case around 1.0005) would fix the opposite offset pattern, where the larger wavenumbers are further away from those predicted. This rough 1.0005 ratio is also only for the 49-0 and is completely different from the ratio needed to fix the other large error which is the 43-0 band. Band-by-band adjustments were attempted and did reduce the number of bad lines in the MARVEL program, but due to the lack of any physical justification for why some bands needed certain adjustments, the 49-1 and 43-0 transitions were simply deleted and the other bands had their uncertainties increased. Further analysis could be done to correct the measurements, but the bands involved are already well characterised by other sources.

3.2.3 72ByHeKiLe [92]

This source provides 9 lines from laser flourescence of the B-X system terminating on the $v = 13$ lower vibrational state. The measurements given are wavelengths in air and therefore were converted to vacuum wavelengths before then converting them to wavenumbers in cm^{-1}. The vacuum wavelength is given by

$$\lambda_{\text{vacuum}} = \lambda_{\text{air}} \times n$$

where n is the refractive index of air. The paper stated the experiment was conducted at room temperature, but without mention of the pressure but standard atmospheric pressure was assumed. The refractive index used to convert the wavelengths was $1.000\,271\,567$, calculated using the NIST webtool [108]. A -0.5 Å offset recommended by the authors was applied before the refractive index adjustment. The predicted MARVEL energy levels were still around $1\,\mathrm{cm}^{-1}$ off from expected and so this source was not included in the final run.

3.2.4 77SiHo [7]

This is an extensive B-X spectrum from $18\,000\,\mathrm{cm}^{-1}$ to $19\,000\,\mathrm{cm}^{-1}$. A calibration correction to this source recommended by Luc and Gerstenkorn was applied [109]. According to this study, the 77SiHo wavenumbers are too high by $0.012\,\mathrm{cm}^{-1}$. This should apparently reduce the uncertainty from $0.015\,\mathrm{cm}^{-1}$ to $0.008\,\mathrm{cm}^{-1}$. The change in uncertainty was not applied because the estimated uncertainties produced by MARVEL suggested that the original value was already an underestimate.

3.2.5 79BrCrHa [96]

A 1979 study by Brand, Cross, and Hayward [96] is of note because a photograph of their spectrum is provided, and the polarisation labelling technique used to select certain transitions demonstrates of the density of some portions of the iodine spectrum. During the MARVEL process it became clear that four of the transitions given in the table of the study were misassigned. The $P(15)$, $R(15)$, $P(17)$, and $R(17)$ lines visible in the middle of Fig. 9 are given as belonging to the $35 - 0$ band whereas the labelled photograph of the spectrum shown in the Fig. 9 gives these as belonging to the $34 - 0$ band. The table assignments must be incorrect and not the photographed spectrum due to the fact that transitions starting from $J = 15$ and $J = 17$ should not lie between transitions starting from $J = 60$ and $J = 71$ of the same vibrational band as is implied by the provided assignments and can be seen in the figure.

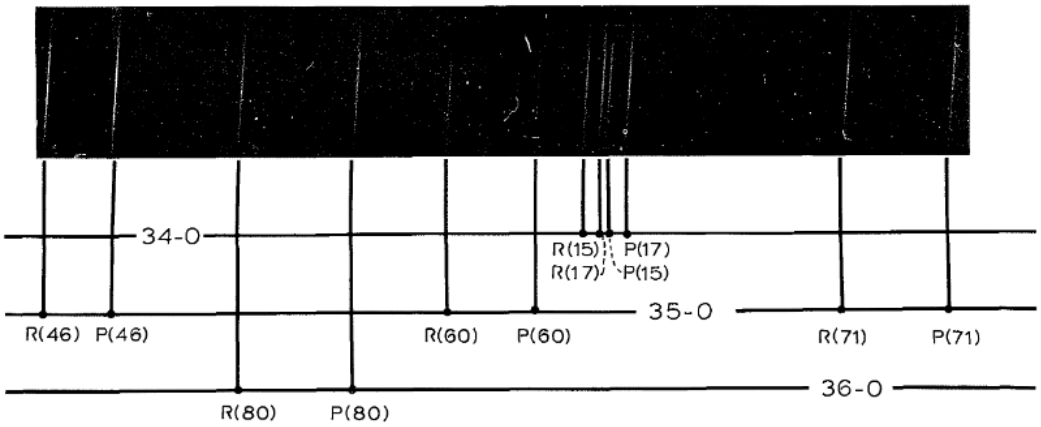


Figure 9: Portion of the spectrum of 79BrCrHa on a photographed plate.

3.2.6 80TeMcSu [99]

[99] The absolute numbering of the D state vibrational levels was not known, so the vibrational assignments given were relative to a $v' = n$ upper vibrational state. Tellinghuisen in a 1983 study [110] was able to assign a cluster of D-X fluorescence transitions originating from $v' = 0 - 3$, the first time any absolute assignments for the D state were found. This study was not rotationally resolved however. According to Bartels *et al.* the n in this study corresponds to $v' = 203$ [87]. The approximate band position expected from transitions in the Tellinghuisen paper (80TeMcSu) can be calculated from the $v - 0$ upper state progression band origins measured by Hoy and Lipson [88] alongside the high v vibrational level energies of the ground state calculated in 80TeMcSu [99] to double check that n is indeed 203.

3.2.7 93IsTaOb [72]

The ion-pair state is labelled $1_g(^1\mathrm{D})$ as it correlates with the $^1\mathrm{D}_2$ I^+ dissociation limit. This state is not to be confused with the $D0_u^+$ state which has direct single-photon transitions with the $X0_u^+$ ground state and has rotationally resolved spectra included in the MARVEL analysis (source 80TeMcSu [99]), or the $D'2_g$ state [3]. The f sublevels can

[3]Due to the favourability of $\Delta\Omega = 0$ ion-pair valence transitions, the $D'2_g$ state and the $A'2_u$ spin component of the $^3\Pi$ first excited state form a well studied system. Spectra of this system in the literature are either not rotationally resolved [111] or not provided in the paper [112–114].

be inferred when there is a Q branch transition at either the initial, pump transition or the second, probe transition. The rotationless parity is always *e* for the X state. If the intermediate state is $B0_u^+$ then a 1_g e level is accessed if the second, probe, transition is in the Q branch and a 1_g f is accessed if the probe transition is in the P or R branch. When the $A1_u^+$ is used as intermediate then the the initial pump transition determines the final rotationless parity, as the transitions measured from the $A1_u^+$ state are either the P or R branch as $\Delta\Omega = 0$ for $A1_u^+ \rightarrow 1_g$. The second transition in this case takes *e* to *e* and f to f based on the rotationless parity selection rules.

3.2.8 96ApLeBeGe [69]

These transitions were measured by the researchers at the Aimé Cotton along with the main B-X atlas but was only assigned for this study. The lines and assignments were kindly provided by Peter Bernath [115]. [69]

3.2.9 Hyperfine Sources

The spectroscopic studies of iodine for the purpose of laser and clock frequency calibration resolve the hyperfine structure of B-X transitions, caused by a change in nuclear spin. A considerable number of the hyperfine resolved studies measure only the relative frequencies of each component and not an absolute frequency, unless these could be linked to an absolute frequency they could not be included in the network. The approximate centre of gravity of a B-X rovibronic line is 5/9 the distance between the "*t*" and "*b*" hyperfine components, corresponding to the first and last components for even-J or the second and second-to-last hyperfine components for odd-J [116]. An odd-J B-X rovibronic line has 21 components and an even-J line has 15 [116]. When the relevant measurements were available, this was the method used for the conversion of hyperfine data to rovibronic line positions. As an example, a 2016 study of four iodine lines at $578\,\mathrm{nm}$, (source 16KoAkHoIn), the authors measured the hyperfine line positions relative to the first component, and measured the absolute frequency of the first component, shown as a_1 in Fig: 10. To get a line centre based on the 5/9ths rule from the data provided it was a case of first calculating the absolute frequencies of the a_2 and a_{20} components using the relative frequencies provided. Then finding the point 5/9ths the distance between the components. For the P33 (16-1) line, the line centre is

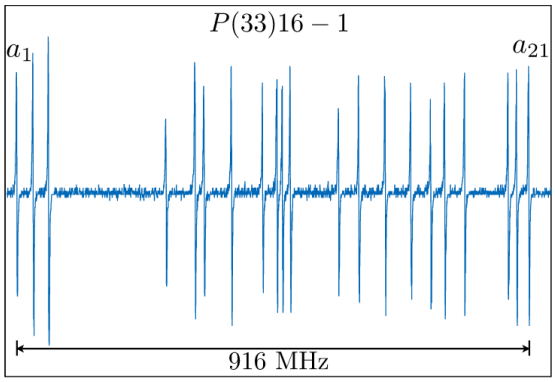


Figure 10: An example of the hyperfine resolved spectrum of a rovibronic line from source 16KoAkHoIn [52]

then

$$\begin{aligned}
\sigma_{a1} &= 518\,287\,542\,790 \pm 7\,\mathrm{kHz} \\
\sigma_{a2} &= \sigma_{a1} + 29\,346.9 \pm 2.0\,\mathrm{kHz} \\
\sigma_{a20} &= \sigma_{a1} + 893\,989.8 \pm 2.0\,\mathrm{kHz} \\
\sigma_{centre} &= \sigma_{a2} + \frac{5}{9}(\sigma_{a20} - \sigma_{a2}) \\
&= 518\,288\,052\,494.0 \pm 7.5\,\mathrm{kHz}
\end{aligned}$$

where the $7.5\,\mathrm{kHz}$ uncertainty was calculated by propagating the provided uncertainties of the hyperfine components through the equations used. This is almost certainly an underestimate as the final frequency is an estimate of the line-centre based on and should not necessarily correspond the uncertainties of each measured hyperfine component. The choice of uncertainty to give to MARVEL for these measurements was $20\,\mathrm{MHz}$ based on the line profiles from [109].

For studies where a few hyperfine lines were given which are not the transitions necessary to calculate the resulting line centre from the 5/9 rule, the mean average of the hyperfine components was taken. In this case an uncertainty of $50\,\mathrm{MHz}$ was used. Where only single components are available they were used if between and including components 5 and 10 for even-J, or between and including 7 and 14 for odd-J (corresponding to the middle third). For these a $100\,\mathrm{MHz}$ uncertainty is used. Despite

the large uncertainties relative to the uncertainties of the hyperfine measurements (usually less than $100\,\mathrm{kHz}$, the superfluous significant figures are kept. The uncertainties are explicit in the MARVEL input so this should not be misleading as long as the uncertainty of the measurement is not inferred from the number of significant figures given in the input file.

A 1997 study by Sansonetti [25] (source 97Sansonet) measured a hyperfine component of 102 B-X rovibronic transitions, 101 of which were transitions present in the Aimé Cotton atlas. All but four of the transitions provided are the "*t* " hyperfine component, which is the first or second component for an even- or odd-J line respectively. The wavenumbers of the *t* component are lower than the measured line centres from the atlas by around $0.01\,\mathrm{cm}^{-1}$ which is greater than the $0.002\,\mathrm{cm}^{-1}$ accuracy of the line centre values. The four other lines are the *i* and *r* components. The *i* component is the 10th in a 15 component even-J transition, or the 13th component for odd-J. The *i* component wavenumbers differ from the corresponding Aimé Cotton atlas wavenumbers by around $0.003\,\mathrm{cm}^{-1}$, and fit the criteria laid out previously to determine proximity to the line centre, so are included in the MARVEL analysis with a stated uncertainty of $0.004\,\mathrm{cm}^{-1}$. The singular *r* component measured, being the fourth component out of 21, is included due to the transition in question not appearing in the Aimé Cotton atlas.

4 Conclusion and Further Prospects

The aim of this project, to produce an extensive and self consistent network of energy levels with associated uncertainties using the available data, has been met. The scope of this task was larger than anticipated due to the format of the data. Now that the transitions have been compiled into one place and in a digital form, adding to and refining this database should be straightforward.

The fact that the MARVEL process requires a uniform set of energy levels poses some issues for iodine due to the wide range of resolution in the literature. The choice to use rotation-vibration levels means that the final MARVEL energy levels contain no hyperfine structure despite the availability of some hyperfine resolved transitions for the B-X system. It also means the energy levels of entire electronic states are omitted due to unresolved rotational structure. There exist studies of the $\beta 1_g$ state [117], and the

$D'2_g$ and $A'2_u$ [114] states providing only vibrational bands. The only way to include all these data is to form separate networks and perform a MARVEL-like process on each one.

There is a wealth of unnasigned high-resolution iodine spectra. Many iodine studies do not give assignments as these are often not necessary for the calibration tasks which motivate these investigations in the first place. All that is required for calibration are line positions and relative intensities, such as in refs. [10, 12, 13], or even spectra without explicitly determine peaks, refs. [9, 11]. An obvious next step to improving the marvel network is to use peak finding routines to determine line positions from the latter papers and assign these. These are still all B-X absorption spectra however, so the issues with lower vibrational levels of the B state, higher vibrational levels of the X state, and a lack of rotationally resolved data for other states would not be solved. More fluorescence spectrum data as discussed earlier would fill out the missing B and X levels. Another element missing from the existing data are absolute intensity values, the sources providing intensities only give them relative to an arbitrary maximum. If knowledge about transition intensities is required for modelling purposes, it may be necessary to use calculated electronic dipole moments, combined with Frank-Condon and Hönl-London factors to predict intensities [66].

It is hoped that this MARVEL database will find use in eventually producing a comprehensive iodine line list to be part of the ExoMol database [118], and that this is of use in the modelling of iodine plasmas within electric propulsion systems.

Appendix: Further notes on sources

72WiKea [76] and **72KiBeb** [93]: A resonance Raman study of iodine vapour using $4880\,\text{Å}$ and $5145\,\text{Å}$ lasers respectively, to obtain Stokes and Anti-Stokes measurements of the same rovibrational transitions. Transitions assigned to the S branch are band heads and not fully rotationally resolved lines so these are not included. The Stokes and Anti-Stokes measurements of the same rovibrational line differ by $1\,\text{cm}^{-1}$ in one case and by up to $0.5\,\text{cm}^{-1}$ for the rest. The stated uncertainties are either $0.5\,\text{cm}^{-1}$ or $1\,\text{cm}^{-1}$ which matches this discrepancy. The Q0(7-4) line is incorrectly assigned and should be assigned to the 7-3 band instead. Similarly the line labelled Q(0)14-2 should be 14-3 instead. These were likely just misprints in the original paper as the lines were correctly in the $\Delta v = 4$ and $\Delta v = 11$ sections respectively.

75PaDwHo [94]: A fluorescence B-X study using excitation from an argon laser. The lines were assigned using the parameters of Wei and Tellinghuisen, but Tellinghuisen noticed a mistake in these assignments but a reassignment of the measured lines was not provided, instead the correct line positions of the incorrect assignments were [119]. Lines which accessed new levels and were consistent with the rest of the network were included.

77GeLuPe [95]: A single B-X band was recorded and rotationally assigned with reference to a uranium standard. The scan available is quite faded but everything could be discerned.

79TaDaGi [98]: A comparison of a third-harmonic generation spectrum and a photoionisation spectrum at the same frequency. The photoionisation spectrum involves the $D0^+$ state and is reassigned by Brand and Hoy [103]. The tag for this reassignment, 87BrHo, is used. The original authors proposed their THG spectrum was the $E0^+$ state but Brand and Hoy suggest it could only be consistent with $F0^+$. The resonances detected would need to be reassigned however, as this change in assignment also changes the resonance from being from the ground $X0^+$ state to the excited $B0^+$ state, meaning a reassignment is not as simple as changing the electronic state label of the upper state. The two photon transitions assigned to T-X where T was the unknown excited state corresponded to $18\,000\,\text{cm}^{-1}$ but it can be seen in Fig. 1 that there is no electronic state near $36\,000\,\text{cm}^{-1}$ above the ground state.

79CuLiFrGr [97]: two-photon transitions between the B and E states wavelengths in Å are provided and needed to be converted, and only measured to five significant figures.

83HlFa [16]: Sansonetti noted that the hyperfine component labelled "o" in this work corresponds to the component labelled "t" in most work using letter labels. The o lines being the first or second hyperfine components are not included in the MARVEL process due to being insufficiently close to the rovibronic line centre. The P(123) 15-1 line is in error according to Sansonnetti [25]

85DiPaKhAl, 86DiPaStKh [101, 120]: Papers with the same data. A Coherent anti-Stoke Raman scheme is used to produce a few transitions within the X state. Only a few lines and not very high resolution as the study was a proof of concept of the technique rather than an iodine investigation.

86IsToShTa [102]: A three-photon study of the $F'0_u^+$ state. The wavenumbers given correspond to the second single-colour two-photon transition from the $B0_u^+$ state, hence the allowed $g \leftrightarrow g$ transition.

92SuMaKl [104]: 34 resolved and assigned transitions in a $2\,\text{cm}$ region. Note that the bands are given in a reversed from standard order, as (v'', v') instead of (v', v'')

94KrBoKnTi [22]: A hyperfine study of the R(92) 0-15 line. The absolute wavenumber used is simply taken and adjusted from the infrared portion of the Aimé Cotton atlas. It was unassigned in the original source, and so is used in this study.

04Kish [121]: A fluorescence study to test a method of detecting the $^{129}I_2$ isotopologue. Measured fluorescence data is not provided and the exciting absorption transitions are calculated. Note that from including the absorption wavenumbers into the

MARVEL network during testing, the stated cm^{-1} wavenumbers appear to have been converted from a wavelength of frequency that has not been adjusted for the refractive index of air. This makes sense within the study as it relates to atmospheric detection, but it is not standard practice for energy in wavenumber units.

05Chen [40]: Contains relative hyperfine measurements for 79 B-X lines, one of which has an absolute measurement. Only the line for which an absolute measurement was provided was used in the MARVEL analysis.

10Williams [106]: Fluorescence spectra excited by 5 different lasers. Covers the much needed higher vibrational levels of the ground state. The transitions were in supplementary data kindly sent by the author [122].

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